

DAX CHALLENGE 2026*Data–AI–Experience Challenge — Integrating Data and AI for a Sustainable Pahang***TECHNICAL REPORT TEMPLATE**

Organised by Pusat Sains Matematik (PSM) and MathElites, UMPSA

Field	Team Entry
Project Title	NADA: New-Era Assistance Distribution Agility
Selected Sub-theme / Sector	Socioeconomic
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Executive Summary

Low unemployment does not always mean low economic hardship. Analysis of Department of Statistics Malaysia (DOSM) district-level data shows that, in Pahang, employment and poverty have decoupled: the districts with the most jobs are not the districts with the least poverty. This project addresses that gap across Pahang's 11 administrative districts, where the state's only direct measure of household hardship, the Household Income and Expenditure Survey (HIES), is released just once every two to three years, leaving planners with frequent employment data but no current picture of who is actually struggling.

Using DOSM's OpenDOSM labour-force, poverty, amenities, demographic and inflation datasets, the project first tested statistically whether unemployment reflects household poverty. Pooled correlation across 33 district-year observations found no defensible relationship between unemployment and poverty (Pearson $r = -0.02$, all $p > 0.8$; OLS $R^2 \approx 0.000$), while Labour Force Participation Rate showed a consistent, significant negative relationship ($r = -0.56$, $p = 0.0006$; $R^2 = 0.32$). Lipis and Maran combine below-median unemployment with above-median poverty and below-average participation, consistent with a working-poor pattern rather than genuine labour-market health.

Building on this, a District Vulnerability Index (DVI) was constructed from four IPCC-based dimensions and validated against actual poverty ($r = 0.567$, $p < 0.001$), improving from unreliable in 2020 to good by 2024; Bera, Lipis and Maran emerged as the most persistently vulnerable districts. A nowcasting engine, trained on six years of HIES and PCHIP-completed data and validated by leave-one-out cross-validation, estimates median household income ($R^2 = 0.66$, MAPE = 6.31%) more reliably than poverty ($R^2 = 0.53$, MAPE = 48.07%); its 2025–2027 projections flag Bera, Pekan, Jerantut, Lipis and Rompin as districts of greatest concern, and show that rising income does not always mean falling poverty.

An unsupervised clustering layer, cross-validated against Ward hierarchical clustering, groups districts by shared vulnerability type rather than severity, revealing that Lipis carries a compound labour-and-living-condition risk the composite ranking alone does not surface.

These findings are delivered through NADA, a three-page Power BI dashboard mapping directly to the project's objectives, giving UPEN Pahang, district councils and NGOs a continuously updatable, clearly labelled complement to official HIES statistics for prioritising districts between survey rounds.

1. Introduction

1.1 Background

Malaysia's principal source of household income and poverty statistics, HIES, is conducted periodically, while district-level HIES results are available for Pahang for 2019, 2022 and 2024. This creates a structural information gap: annual employment and socioeconomic indicators are available more frequently than direct district-level poverty and household-income observations.

A common planning assumption is that employment conditions can stand in for household wellbeing: if unemployment is low, hardship should also be low. However, Pahang data do not support that assumption. Districts with relatively low unemployment can still record elevated poverty, particularly where labour-force participation is lower or employment does not translate into sufficient household income.

The information gap matters to economically vulnerable households and to institutions responsible for distributing scarce resources. UPEN Pahang, district councils and welfare organisations need a timely way to identify where hardship may be persistent or increasing without treating the latest unemployment rate as a complete measure of welfare.

1.2 Problem Statement

Across the 11 districts of Pahang, household poverty and income are observed much less frequently than employment indicators. Consequently, planners lack a timely, integrated mechanism to track how labour-market conditions, price pressure, demographic dependency, education access and basic-service deficits may combine into district-level vulnerability between HIES rounds.

The central analytical gap is therefore not the absence of data, but the absence of a validated way to combine more frequently available indicators into an interim vulnerability signal. NADA addresses this gap by testing the unemployment–poverty relationship, constructing a transparent DVI, and developing a nowcasting approach that explicitly distinguishes estimated values from official HIES observations.

1.3 Research Questions

To address the policy information gap in Pahang and systematically evaluate district-level economic vulnerability, this project poses five logically progressing research questions:

- **RQ1 (Descriptive - What):** To what extent does headline unemployment decouple from household poverty and multi-factor economic hardship across Pahang's administrative districts?
- **RQ2 (Spatial & Diagnostic - Where):** Which of Pahang's 11 administrative districts experience the greatest composite risk when evaluated using the multi-dimensional District Vulnerability Index (DVI)?
- **RQ3 (Explanatory - Why):** Which specific underlying dimensions are the primary drivers of vulnerability in high-risk districts?
- **RQ4 (Predictive - What Will Happen):** Can regularized fixed-effects panel models accurately nowcast median household income and poverty risk during inter-survey gap years using high-frequency socioeconomic predictors?
- **RQ5 (Prescriptive - What Should We Do):** How can multi-dimensional DVI outputs and nowcasting predictions be integrated into a decision-support system to guide resource allocation, targeted aid, and policy prioritization for UPEN Pahang, district councils, and local NGOs?

1.4 Project Objectives

Objective 1: To quantify the relationship between employment indicators and household poverty across Pahang's 11 administrative districts and demonstrate statistically that unemployment is a poor stand-in for hardship; a district can have almost no unemployment and still have widespread poverty.

Objective 2: To develop a data-driven District Vulnerability Index that classifies districts by household economic risk using indicators available annually or monthly.

Objective 3: To build and validate a model that nowcasts estimates of present-day values before official figures exist for district median income and poverty rate for years in which no HIES survey was conducted.

Objective 4: To deliver an interactive dashboard that allows non-technical users in state government to see current estimated vulnerability by district.

1.5 Scope of Study

Provide placeholders for the boundaries of the project:

Boundary	Project scope
Geography	Pahang, Malaysia; all 11 administrative districts
Period	2019–2024 for HIES-matched analysis; 2020–2024 for the final DVI panel
Analytical observations	55 district-year rows in the 2020–2024 indicator panel; 33 matched HIES district-years
Core variables	Unemployment, LFPR, EPR, dependency ratio, school enrolment, amenities, inflation, poverty and median income
Excluded	Agriculture dependence and physical hazard exposure because consistent annual district-level data were unavailable
Interpretive boundary	DVI is a relative socioeconomic vulnerability signal, not an official poverty headcount

Table 1.1 — Scope of study for NADA Project

1.6 Stakeholder and Decision Context

Primary stakeholders are UPEN Pahang, district councils/local authorities, and NGOs/aid groups. The decision supported is which districts should receive priority for development funds, targeted aid or infrastructure investment in a given year.

2. Dataset Understanding

2.1 Dataset Source

All datasets are open data from DOSM's OpenDOSM portal, supplemented by the Ministry of Education source for contextual variables.

Dataset	Provider	Coverage	Level	Purpose
Labour Force Survey (unemployment, LFPR, EPR)	DOSM / OpenDOSM	2018–2024	District	Decoupling analysis; labour DVI dimension; predictors
HIES – absolute poverty	DOSM / OpenDOSM	2019, 2022, 2024	District	Ground-truth poverty benchmark and target
HIES – median income	DOSM / OpenDOSM	2019, 2022, 2024	District	Nowcasting target
Basic Amenities	DOSM / OpenDOSM	2016, 2019, 2022, 2024	District	Living-condition vulnerability
Population Estimates	DOSM / OpenDOSM	2018–2025	District	Dependency ratio and denominators
Consumer Price Index	DOSM	2010–Jun 2026	State (Pahang)	Price-pressure dimension
Student enrolment	Ministry of Education	2018–2025	District	Education-access proxy

Table 2.1 — Core datasets used, all sourced from DOSM's OpenDOSM portal and Ministry of Education

2.2 Unit of Observation

One row of the primary analytical dataset represents one Pahang administrative district in one calendar year, carrying labour-force, demographic, amenities, school and inflation indicators. The HIES poverty table uses the same district-year grain but is populated only in HIES release years.

2.3 Data Dictionary

Variable	Description	Unit	Role
district	Standardised Pahang district name	—	Identifier/geographic
year	Calendar year	Year	Temporal
unemployment_rate_pct	Unemployed persons as % of labour force	%	Predictor; DVI
participation_rate_pct (LFPR)	Labour force as % of working-age population	%	Predictor; DVI
employment_population_ratio_pct (EPR)	Employed persons as % of working-age population	%	DVI
dependency_ratio	$(\text{Age } 0\text{--}14 + 65+) / \text{age } 15\text{--}64 \times 100$	Ratio	DVI
school_enrolment_rate	Students / population aged 5–19 $\times 100$	%	Predictor; DVI
piped_water / sanitation / electricity	Household access share	%	DVI; reversed to deprivation
pahang_inflation_yoy_pct	Pahang state year-on-year CPI change	%	DVI price pressure
poverty_absolute_pct	Absolute poverty incidence (HIES)	%	Target / validation
median_income	Median household income (HIES)	RM	Target

Table 2.2 — Data dictionary for the variables actually used in all analytical tasks

2.4 Temporal and Geographic Coverage

- Earliest source coverage extends to 2016 for population/amenities; the assembled DVI indicator panel begins in 2020.
- Latest district-level socioeconomic/HIES coverage is 2024; CPI extends to June 2026 at Pahang state level.
- All 11 Pahang districts are represented in the core datasets.
- District-level data are used for labour, amenities, population and HIES; CPI is state-level and applied uniformly as the best available price-pressure proxy.

2.5 Dataset Limitations

- Only three district-level HIES release years are available for validation.
- Publication frequencies differ across monthly CPI, annual labour indicators and periodic HIES/amenities surveys.
- District aggregation can mask within-district variation.
- State-level CPI cannot capture district-specific price differences.
- Some contextual variables, such as enrolment, are proxies rather than direct household-welfare measures.

3. Data Preparation and Preprocessing

The data preparation workflow was designed to integrate multiple DOSM/OpenDOSM and Ministry of Education datasets into a consistent district-year panel for Pahang's 11 administrative districts. The preprocessing consisted

of raw-data validation, geographic and temporal standardisation, numeric conversion, duplicate checking, range validation, district-year aggregation, missing-value assessment, feature construction and final analytical-dataset preparation. The final DVI analytical dataset contains 55 district-year observations covering 11 districts from 2020 to 2024.

3.1 Data Cleaning Workflow

The preprocessing pipeline integrates DOSM/OpenDOSM and Ministry of Education datasets into a consistent district-year panel. Each source was checked before integration, then standardised and aggregated to the common analytical grain.

Data issue	Detection	Treatment	Justification
Missing observations	isna() counts and percentages	Retain structural missingness; interpolate selected annual variables later	Missingness mainly reflects source-year coverage
Exact duplicates	duplicated()	None removed	No exact duplicates detected
District-year duplicates	district + year key	None removed	One analytical row must represent one district-year
District-name inconsistency	Cross-dataset comparison	District crosswalk	Prevents failed joins and false missingness
Date/year inconsistency	Date/year validation	Common calendar-year key	Required for annual merging
Numeric formatting	Data-type/special-character checks	Numeric conversion	Enables calculation and modelling
Invalid values	Range/sign checks	Investigate and exclude invalid core observations	Maintains theoretical validity
Coverage mismatch	Year/district coverage checks	Merge at common grain; retain coverage flags	Different source publication schedules

Table 3.1 — Data workflow in data cleaning

The preprocessing pipeline explicitly completed raw-data processing, district crosswalking, district standardisation, date standardisation, numeric conversion, missingness analysis, duplicate checks, unit/range verification, cross-dataset validation, temporal validation, district-year aggregation and integrated master-panel construction.

The initial master district-year panel was constructed for all 11 Pahang administrative districts from 2020–2024, giving 55 district-year observations with no duplicate district-year keys. This panel formed the basis of the DVI analysis, as the core district-level indicator panel was available consistently from 2020 onward. Individual datasets were subsequently merged into this common structure, with each source retaining its original temporal coverage. Labour-force data covered 2018–2024; HIES income and poverty data covered 2019, 2022 and 2024; population data comprised Pahang-level estimates for 2018–2019 and district-level estimates for 2020–2025; amenities covered 2016, 2019, 2022 and 2024; and student enrolment covered 2017–2025. Missingness was therefore treated primarily as a coverage issue rather than assuming that an absent observation represented zero.

For Objective 3, the analytical panel was subsequently extended to 2019 to incorporate the earliest available district-level HIES observation as a historical anchor for the nowcasting procedure. This produced a six-year 2019–2024 panel used for PCHIP completion and predictive modelling. The 2019 observation was therefore

added specifically to support the nowcasting methodology, while the DVI analysis remained based on the 2020–2024 panel.

3.2 Missing Values

The initial 2020–2024 master panel contained 55 district-year observations, with structural missingness mainly in HIES and amenities variables due to their non-annual publication. For Objective 3, the panel was subsequently extended to 2019–2024 to incorporate the 2019 HIES observation as an additional historical anchor, giving 66 observations. Missingness was therefore treated as a coverage issue rather than zero. For nowcasting, the observed HIES years (2019, 2022 and 2024) were retained as actual values, while 2020, 2021 and 2023 were completed using PCHIP interpolation and flagged accordingly. Thus, each target variable contains 33 observed and 33 interpolated historical district-year observations before the 2025–2027 nowcast period.

3.3 Duplicate Records

- Exact duplicates were assessed across complete rows; none were found in the validated source datasets.
- Analytical duplicates were assessed using district + year; the master skeleton and integrated panel contained zero duplicate district-year keys.
- No observations were removed specifically because of duplication.

3.4 Outlier Detection

Potential extremes were evaluated using descriptive statistics, minimum/maximum checks and domain validity rather than automatic deletion. Labour-force participation ranged from 52.7% to 73.0%, unemployment from 1.2% to 4.0%, and EPR from approximately 50.7% to 70%. These were considered plausible and retained.

Amenities also remained within plausible ranges: piped-water access 84.3%–100.0%, sanitation 91.1%–100.0%, and electricity 93.5%–100.0%. The population contained no negative values, and crop production ranged from zero to 117,093 tonnes in the 2017–2022 source. Extreme but plausible observations were retained because a genuine socioeconomic extreme may itself be analytically meaningful.

3.5 Categorical Standardisation and Unit Standardisation

- District crosswalk: “Cameron Highland” → “Cameron Highlands”; “Kuala Lipis” → “Lipis” for the final reporting convention.
- Labour-force and population quantities → thousand persons (000).
- Income and median income → RM; poverty and labour indicators → percentage (%).
- Water, sanitation and electricity access → percentage (%); crop production → tonnes; inflation → percentage (%).
- Special characters such as commas and percentage signs were checked before numeric conversion.

3.6 Feature Engineering

Derived feature	Formula / logic	Purpose
Dependency ratio	$(\text{Population } 0\text{--}14 + 65+) / \text{Population } 15\text{--}64 \times 100$	Demographic burden on working-age population
School enrolment rate	$\text{students_total} / \text{population aged } 5\text{--}19 \times 100$	Education-access proxy
Piped-water deprivation	$100 - \text{piped-water access \%}$	Aligns direction with vulnerability
Sanitation deprivation	$100 - \text{sanitation access \%}$	Measures lack of adequate sanitation

Electricity deprivation	100 – electricity access %	Measures electricity-access deprivation
Pahang YoY inflation	Annual/monthly CPI transformed to YoY %	Represents state-level price pressure

Table 3.2 — Feature engineering of the dataset to derive meaningful related variables

For the DVI, variables that were not sufficiently available annually were excluded from the final core indicator set. In particular, direct poverty and income measures were retained for validation/target purposes rather than being treated as DVI input indicators.

3.7 Final Analytical Dataset

After cleaning, standardisation and feature preparation, the DVI dataset was finalised for 2020–2024, comprising 11 districts and 55 district-year observations. For Objective 3, the panel was subsequently extended to 2019–2024 to support the nowcasting analysis. The dataset retains labour, population, amenities, enrolment and inflation indicators, while income and poverty are treated as target/validation variables, with observed/interpolated flags preserved for modelling.

3.8 Data-processing flow

Raw DOSM/MOE datasets → data validation → missing/duplicate checks → district and year standardisation → numeric/unit harmonisation → district-year aggregation → coverage validation → feature engineering and observed/interpolated flags → 2020–2024 DVI dataset → 2019–2024 extended panel for Objective 3.

4. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to identify the major patterns, differences and relationships within the Pahang district-level socioeconomic data. The analysis was designed to address the relationship between employment and poverty, differences between districts, temporal patterns, and the suitability of annually available indicators as proxies for household economic vulnerability. The EDA follows five complementary perspectives: descriptive, temporal, geographic, group comparison and relationship analysis.

4.1 Descriptive Analysis

The primary DVI panel contains 55 district-year observations from 2020–2024; the poverty-validation dataset contains 33 observations from the three HIES release years. Across the 33 HIES observations, absolute poverty has a mean of 6.22%, standard deviation of 3.13 percentage points, median 6.4%, minimum 0.4% and maximum 12.0%. Relative poverty has a mean of 7.24% and maximum of 18.5%.

Poverty rates vary considerably across districts, with absolute poverty ranging from 0.4% to 12.0% in the HIES observations. The substantial spread indicates that socioeconomic hardship is unevenly distributed across Pahang. State-level planning should account for district-level differences rather than applying uniform interventions across all 11 districts.

4.2 Temporal Analysis

Labour-force data provide annual observations from 2018–2024, while poverty and income are only observed in 2019, 2022 and 2024. The 2020 period shows a visible labour-market disruption: Bentong's labour force declined from 60.3 thousand in 2019 to about 52.5 thousand in 2020 before recovering to 57.4 thousand in 2024; Bera declined from 43.3 thousand to 35.2 thousand before recovering to 37.8 thousand.

Pahang's overall CPI inflation also changed sharply: approximately -1.18% in 2020, around 3.0% in 2021 and 2022, then 2.68% in 2023 and 2.45% in 2024. The temporal pattern shows that vulnerability can shift with macroeconomic conditions even after headline employment indicators recover. An annually updated vulnerability measure should incorporate more than unemployment alone and should account for changing labour-market, living-condition and price-pressure conditions.

4.3 Geographic Analysis

District comparisons reveal clear differences. Lipis and Maran have below-median unemployment but above-median poverty and below-average LFPR. The DVI later identifies Maran, Lipis and Jerantut as high-vulnerability districts, while Cameron Highlands, Bentong and Kuantan are among the lowest.

Therefore, low unemployment does not necessarily indicate strong household economic conditions. A district may have high employment participation while households remain economically vulnerable because of low-productivity or low-income employment. UPEN Pahang and district authorities should avoid using unemployment rankings as the sole basis for identifying districts requiring socioeconomic assistance.

4.4 Group Comparison

The comparison between districts based on unemployment and poverty revealed a particularly important group: districts with low unemployment but relatively high poverty. Lipis and Maran fall within this potential "working-poor" group. This demonstrates that employment status alone cannot distinguish economically secure districts from districts where residents may be employed but remain financially vulnerable.

The EDA also supports a multidimensional interpretation of vulnerability. The DVI subsequently groups indicators into four dimensions: labour-market vulnerability, household economic proxy, living-condition vulnerability and macroeconomic price pressure. This approach recognises that different districts may experience hardship through different mechanisms rather than through a single common factor.

Districts vary in *which* dimension drives their vulnerability; some are more strongly associated with labour-market conditions, others with living-condition or demographic pressures. However, this variation is not unique to each district: several districts share closely similar vulnerability profiles despite differing in overall severity, suggesting that districts can be grouped by the type of hardship they face rather than treated as eleven isolated cases. Interventions should be tailored to the dominant vulnerability dimension shared by each group of districts, rather than applying either a single state-wide intervention or eleven entirely separate ones.

4.5 Relationship Analysis

Relationship	Result	Interpretation
Unemployment ↔ absolute poverty	Pearson $r = -0.020$; Spearman $\rho = -0.038$; Kendall $\tau = -0.021$; $p > 0.8$	No statistically defensible pooled relationship
LFPR ↔ absolute poverty	Pearson $r = -0.564$, $p = 0.0006$; OLS $R^2 \approx 0.32$	Higher participation is associated with lower poverty
Mean income ↔ poverty	$r = -0.614$	Higher income tends to accompany lower poverty
Median income ↔ poverty	$r = -0.501$	Moderate negative association
LFPR ↔ Employment Population Ratio	$r = 0.998$	Strong redundancy among labour indicators

Labour force ↔ employed population	$r = 1.000$	Near-perfect redundancy
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Table 4.1 — Relationship analysis of the variables

4.6 Key EDA Insights

The EDA produced four major findings that directly informed the subsequent DVI methodology.

Finding 1 — Unemployment is not a reliable standalone measure of poverty

The unemployment–poverty relationship was effectively zero in the pooled sample ($r=-0.02$, $p=-0.04$, $R^2\approx 0$). Districts with low unemployment do not consistently have low poverty. Headline unemployment does not adequately capture household economic hardship in Pahang. Welfare and development targeting should incorporate multidimensional indicators rather than unemployment alone.

Finding 2 — Labour-force participation provides a stronger signal

LFPR had a statistically significant negative relationship with absolute poverty ($r=-0.564$, $p = 0.0006$). Higher LFPR is associated with lower observed poverty. The ability and willingness of the working-age population to participate in the labour market may provide more useful information about local economic vulnerability than unemployment alone. LFPR is retained as an important component of the DVI.

Finding 3 — Lipis and Maran illustrate the "working-poor" problem

Lipis and Maran were identified as districts with below-median unemployment but above-median poverty, while also showing below-average labour-force participation. Relatively low unemployment coexists with elevated poverty in these districts. Employment statistics can conceal economic insecurity when employment is low-productivity, low-paid, informal or insufficient to lift households above the poverty line. These districts warrant closer investigation and demonstrate the need for a vulnerability measure that goes beyond unemployment.

Finding 4 — Vulnerability is multidimensional and district-specific

The EDA showed substantial differences between districts and strong correlations among some indicators. The resulting DVI therefore combines multiple dimensions rather than treating one indicator as a complete representation of socioeconomic wellbeing. Districts exhibit different combinations of labour-market, demographic, living-condition and price-pressure characteristics. No single socioeconomic indicator sufficiently represents vulnerability across all districts. A transparent, multidimensional DVI is more appropriate for district-level decision support than an unemployment-only approach.

5. Statistical / AI Methodology

5.1 Analytical Strategy

The project follows a Descriptive → Diagnostic → Predictive → Prescriptive progression. Objective 1 is diagnostic: it tests whether unemployment is a defensible poverty proxy. Objective 2 is descriptive-diagnostic: it constructs and validates a DVI. Objective 3 is predictive: it nowcasts income and poverty across the inter-survey gap. The dashboard then translates the analytical outputs into decision support.

5.2 Method Selection

RQ / Task	Method	Why selected	Output
RQ1: employment–poverty relationship	Pearson, Spearman, Kendall + Shapiro–Wilk	Cross-check linear, monotonic and rank relationships; non-normal employment indicators motivate rank-based emphasis	Correlation + p-value
RQ1: uncertainty	District-cluster bootstrap, 95% CI	Resamples whole districts rather than treating repeated district-years as independent	Confidence interval
RQ1: effect size	OLS	Interpretable coefficient and R^2 alongside correlations	Coefficient, p, R^2
RQ2: district disconnect	Median-based quadrant classification	Directly operationalises low-unemployment/high-poverty claim	District quadrant
RQ3: DVI	Min-max/z-score normalisation + equal weighting	Transparent and reproducible for a 55-row panel	DVI score 0–1
RQ3: weighting check	PCA	Benchmarks equal weighting against variance-maximising alternative	PC1 variance/rank correlation
RQ4: gap completion	PCHIP	Shape-preserving interpolation between sparse HIES anchors	Completed series + flags
RQ4: poverty	OLS, Ridge $\alpha=10$, depth-3 tree, Decision Tree	Compares baseline, regularised and tree alternatives	LOOCV metrics
RQ4: income	Same four candidates as poverty	Selects best-performing interpretable candidate	LOOCV metrics
RQ4: uncertainty	5,000 district-cluster bootstrap + 5,000 Monte Carlo draws	Propagates grouped and predictor uncertainty	95% prediction intervals
RQ5: decision support	Power BI + planned database-grounded AI assistant	Connects validated outputs to operational decisions	Dashboard/query outputs

Table 5.1 — Method-selection matrix linking every research question to its analytical task, method and output

5.3 Baseline Model

For the decoupling analysis, unemployment rate itself functions as the implicit "baseline" indicator the project is testing against and LFPR is evaluated as a candidate improvement over it. For the DVI, equal-weighted aggregation is the primary transparent model, with PCA used as a comparison rather than the production formula. For predictive nowcasting, Ridge and Regression Tree are the baseline. This choice prioritises reproducibility and explainability over fitting a small dataset with a more complex weighting scheme.

5.4 Objective 1 Methodology — Testing the Employment–Poverty Relationship

5.4.1 Analytical Procedure

Objective 1 required testing whether unemployment rate reliably proxies household poverty at district level, and identifying where that relationship breaks down.

The 33 district-year observations (11 districts × 2019, 2022 and 2024 HIES years) were screened using Shapiro–Wilk. Unemployment $W = 0.930$, $p = 0.0354$ and LFPR $W = 0.922$, $p = 0.0206$ were non-normal; absolute poverty $W = 0.965$, $p = 0.3751$ was approximately normal. Spearman correlation was therefore designated primary, with Pearson and Kendall as cross-checks.

Indicator	Pearson r (p)	Spearman ρ (p)	Kendall τ (p)
Unemployment	−0.020 (0.910)	−0.038 (0.836)	−0.021 (0.864)
LFPR	−0.564 (0.0006)	−0.532 (0.0014)	−0.374 (0.0024)

Table 5.2 — Pooled correlation of employment indicators against absolute poverty rate, $n = 33$ district-years

District-clustered bootstrap resamples whole districts to respect within-district dependence. The unemployment correlation interval spans −0.475 to 0.377 (bootstrap $p = 0.837$), while LFPR spans −0.775 to −0.047 (bootstrap $p = 0.034$). OLS gives $R^2 \approx 0.000$ for unemployment and $R^2 = 0.318$ for LFPR.

Average unemployment and poverty for 2019/2022/2024 are compared with Pahang medians of 2.43% and 6.80%. This identifies Lipis and Maran as low-unemployment/high-poverty districts. Their mean LFPR across flagged district-years is 60.3%, below the full-sample mean of 63.8%.

District	Avg. unemployment	Avg. poverty	Quadrant
Jerantut	2.90%	9.10%	High unemployment / High poverty
Pekan	2.43%	9.70%	High unemployment / High poverty
Bera	2.80%	6.80%	High unemployment / High poverty
Rompin	2.47%	8.97%	High unemployment / High poverty
Lipis	1.83%	8.07%	Low unemployment / High poverty
Maran	2.13%	7.70%	Low unemployment / High poverty
Kuantan	2.37%	3.20%	Low unemployment / Low poverty
Bentong	2.27%	2.70%	Low unemployment / Low poverty
Cameron Highlands	1.80%	2.43%	Low unemployment / Low poverty

Table 5.3 — District positioning, unemployment vs. poverty (Pahang-median thresholds)

5.4.2 Model Evaluation

The employment–poverty relationship is evaluated on internal consistency across independent methods rather than on a single hold-out test, since Objective 1 is a diagnostic (relationship-testing) task. A conclusion is only treated as robust here if it holds across all three correlation methods (Pearson, Spearman, Kendall), is confirmed by a bootstrap confidence interval that excludes zero, and is corroborated by OLS regression's coefficient significance and explained variance. By this standard, unemployment fails all four checks simultaneously, while LFPR passes all four convergence across methods, rather than any single p -value, which the report treats as the decisive empirical evidence.

5.4.3 Model Limitations

- $n = 33$ is small, and only 11 districts are independent resampling units.
- Pooling assumes the employment–poverty relationship is reasonably stable across survey years.
- Median-based quadrant labels are threshold-sensitive.
- District aggregates cannot establish household-level causality.

- The low-participation explanation is a hypothesis consistent with the evidence, not a causal finding.

5.5 District Vulnerability Index Model Development

5.5.1 Conceptual Framework

Household economic risk in Pahang is defined, following Abd Rahman & Che Sulaiman (2023) and Siti Hadijah (2012), as financial inadequacy, resource deprivation, or earning instability that prevents a household from maintaining baseline living standards and meeting fundamental human needs. Because income and poverty data are not published annually, the DVI approximates this risk using dimensions built from indicators that are available every year, following the IPCC Exposure–Sensitivity–Adaptive-Capacity framework used in prior Pahang Livelihood Vulnerability Index (LVI) studies (Hahn, Riederer & Foster, 2009).

Adapted (with available data), four-dimension framework:

Dimension	Role	Indicators
Labour-market vulnerability (L)	Employment/income shock exposure and ability to obtain work	Unemployment; reverse LFPR; reverse EPR
Household economic proxy (H)	Indirect household economic pressure	Dependency ratio; reverse school enrolment
Living-condition vulnerability (C)	Basic-service deprivation	Lack of water, sanitation and electricity
Price pressure (P)	Economy-wide exposure	Inflation rate

Table 5.4 — Four dimensions of DVI

The adapted four-dimension DVI is: $DVI(d,t) = [L(d,t) + H(d,t) + C(d,t) + P(t)] / 4$. Each dimension is an arithmetic mean of its surviving indicators, oriented so that higher values indicate higher vulnerability.

5.5.2 Alternative IPCC Pillar Structure

Pillar	Indicators
Exposure (EXP)	Inflation rate
Sensitivity (SEN)	Unemployment; dependency ratio; lack of water/sanitation/electricity
Lack of Adaptive Capacity (LAC)	Reverse EPR; reverse LFPR; reverse school enrolment

Table 5.5 — IPCC-based DVI indicators

The alternative IPCC formulation is $DVI = (EXP + SEN + LAC) / 3$, with the same indicators used. Both formulations use equal averaging to avoid overfitting a 55-observation panel.

5.5.3 Normalisation

All indicators are min-max normalised using pooled minima and maxima across the full district-year dataset so absolute vulnerability levels remain comparable over time. Vulnerability-increasing indicators use $Z = (X - X_{min}) / (X_{max} - X_{min})$; protective indicators are reversed as $Z = (X_{max} - X) / (X_{max} - X_{min})$.

5.5.4 Indicator Screening Against Actual Poverty

Each of the nine candidate indicators was oriented so that higher values imply higher vulnerability. For each district, an indicator was retained only when its correlation with the three matched poverty observations was positive. The same sign-screening was repeated at the Pahang-wide average level. Because each district has only three matched points, the screening rule is used as a directional filter rather than a robust hypothesis test.

District	Indicators retained	Count
Rompin	Inflation, dependency, lack of water/sanitation/electricity, EPR, LFPR	7/9
Lipis	Inflation, lack of water/sanitation/electricity, EPR, LFPR	6/9
Bentong	Unemployment, dependency, EPR, LFPR, school enrolment	5/9
Cameron Highlands	Inflation, lack of water, EPR, LFPR	4/9
Bera	Inflation, lack of sanitation, EPR, LFPR	4/9
Pekan	Inflation, dependency, EPR, LFPR	4/9
Raub	Unemployment, dependency, lack of water, school enrolment	4/9
Temerloh	Inflation, EPR, LFPR, school enrolment	4/9
Jerantut	Unemployment, dependency, school enrolment	3/9
Kuantan	Inflation, LFPR	2/9
Maran	Inflation only	1/9

Table 5.6 – Indicators retained per district after wrong-sign screening (out of 9 candidates), corrected dataset

Maran and Kuantan have only 1–2 surviving indicators, so a single indicator can account for 50–100% of their score. In Bentong, no single indicator exceeds roughly 20%. This is transparent but creates comparability and confidence limitations.

5.5.5 Model Evaluation

Three independent validation approaches were applied to the DVI:

- (i) PCA, to check whether the equal-weight averaging scheme distorts the ranking relative to a data-driven weighting.
- (ii) Correlation against actual absolute poverty rate, the direct external-validity check.

5.5.6 Model Limitations

District-specific formulas are not perfectly comparable because the number and identity of surviving indicators differ. This is particularly important for Maran and Kuantan and motivates confidence flags rather than treating the DVI as an official statistic.

5.6 Statistical / AI Methodology — Objective 3 (Nowcasting)

5.6.1 Historical Completion vs Forecasting

Two stages are deliberately separated. First, PCHIP interpolation fills 2020, 2021 and 2023 between the confirmed HIES anchors of 2019, 2022 and 2024. Second, supervised regression models use the completed 2019–2024 panel to project 2025–2027. Interpolation is mathematical completion of known anchors; forecasting is model-based extrapolation into unknown years.

5.6.2 Interpolation Method: PCHIP

Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) was selected because it preserves monotonicity and avoids the overshoot or oscillation a standard cubic spline can produce from only three anchor points. Interpolated observations are flagged so they are distinguishable from actual HIES values.

5.6.3 Predictive Models

Outcome	Candidates	Selected model	Reason
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Poverty	OLS; Ridge $\alpha=10$; depth-3 Regression Tree; unrestricted Decision Tree	Ridge	Stability and linear interpretability with a small correlated predictor set, despite lower LOOCV score
Median income	OLS; Ridge $\alpha=10$; depth-3 Regression Tree; unrestricted Decision Tree	Depth-3 Regression Tree	Best pooled LOOCV R^2 and lowest MAPE among candidates; splits remain interpretable

Table 5.7 – Predictive model chosen for each of the nowcast variables

5.6.4 Model Development

Multiple correlation measures were considered to assess the strength and consistency of relationships across the candidate indicators. Indicators demonstrating stronger and more consistent associations were retained as predictors for the subsequent modelling stage.

- Poverty target: absolute poverty; predictors: LFPR and school enrolment.
- Income target: median household income; predictors: LFPR, unemployment and dependency ratio.
- No held-out test set was carved out because the panel is small; pooled LOOCV was used across 55 usable district-year rows.
- Hyperparameters were carried over from the team's prior model-selection experiment rather than re-tuned in this run.
- Future predictors for 2025–2027 are extrapolated using district-level linear trends; missing future predictors are imputed by cross-district median.

5.6.5 Predictive Performance

Outcome / model	LOOCV MAE	LOOCV RMSE	MAPE	R^2
Poverty — Ridge	1.80 pp	2.17 pp	48.07%	0.533
Median income — Regression Tree	RM 278.45	RM 356.26	6.31%	0.662

Table 5.8 – Validation performance for each of the nowcasting variables

Poverty's high MAPE partly reflects its small absolute scale: district poverty is roughly 2–13%, so a 1–2 percentage-point miss can become a large percentage error. MAE and RMSE are therefore more informative for poverty. Income's 6.31% MAPE and $R^2 = 0.662$ indicate materially stronger predictive performance.

5.6.6 Explainability

Ridge poverty modelling provides directly readable standardised coefficients and signs, while the depth-3 income tree produces a small set of interpretable splits and leaf-level averages. The project deliberately favours shallow, interpretable models for public-sector decision support.

5.6.7 Uncertainty

Forecast uncertainty for 2025–2027 is generated using district-cluster bootstrap (5,000 resamples) combined with Monte Carlo predictor-noise simulation (5,000 draws). The approach accounts for grouped panel structure and uncertainty in projected predictors. Prediction intervals are reported as 95% intervals; coverage cannot yet be validated because 2025–2027 outcomes are unobserved.

5.6.8 Model Limitations

- Only 55 usable rows provide a thin base for LOOCV, particularly for trees.
- Three of six historical training years are PCHIP interpolations, creating circularity risk and likely optimistic apparent performance.

- Linear predictor extrapolation cannot capture policy shocks, turning points or economic cycles.
- The poverty Decision Tree outperformed Ridge (R^2 0.635 vs 0.533), but Ridge was retained for stability/interpretability.
- The depth-3 income tree creates repeated projected values for some district-years because of shared leaf structure; this is a model artifact, not evidence of true stagnation.

5.7 AI Integration — District Typology via Clustering

5.7.1 Rationale

The DVI (Section 5.5) ranks districts by severity but not by *type* of vulnerability. Given $n = 11$ districts (33–55 district-years), ANN/CNN were rejected as overfitting-prone/inapplicable (no spatial data); K-Means clustering was chosen as the complexity-appropriate method, consistent with Sections 5.4.3/5.5.5/5.6.8.

5.7.2 Method

K-Means was applied to each district's mean dimension profile (L, H, C, P; 2020–2024 average). $k = 2$ gave the best silhouette (0.427), but $k = 3$ was chosen for interpretability, aligning with the existing High/Moderate/Low framing (silhouette = 0.391). To rule out algorithmic artefact, results were cross-checked against Ward hierarchical clustering: ARI = 1.000 (perfect agreement between two independent methods). PCA was used only for 2D visualisation.

5.7.3 Missing data

15 of 44 district-dimension cells were undefined because every indicator in that dimension was screened out for that district (Section 5.5.3) is not an accidental missingness. These were imputed with the Pahang-wide mean (a neutral value) and flagged. Imputed cells get $z = 0$ and are excluded from the dominant-dimension diagnosis, so they can never be mistaken for a real signal. Maran (only 1/4 dimensions survived) is correctly reported as "insufficient data" rather than assigned a fabricated answer.

5.7.4 Evaluation

Each district's dimensions were z-scored against the Pahang mean to find its most disproportionately elevated dimension. Macro/Price Pressure has zero cross-district variance (a state-level figure) and was excluded from this step. It is valid in the composite DVI, but unable to distinguish districts. Cross-tabulating clusters against the existing Vulnerability Level bucket showed only partial overlap, confirming type and severity are complementary, not redundant.

5.7.5 Limitations

With $n = 11$, clusters are indicative typologies, not definitive partitions. $k = 3$ over the silhouette-optimal $k = 2$ is a disclosed interpretability trade-off. Imputed dimensions reduce diagnostic power for low-coverage districts (Bentong, Kuantan).

6. Results and Findings

6.1 Objective 1 Findings — Unemployment vs. Poverty

Finding 1 — Unemployment does not reliably track poverty

Pearson $r = -0.020$ ($p = 0.910$), Spearman $\rho = -0.038$ ($p = 0.836$), Kendall $\tau = -0.021$ ($p = 0.864$), bootstrap 95% CI -0.475 to 0.377 , and OLS $R^2 \approx 0.000$. Unemployment carries essentially no reliable information about which

Pahang districts experience higher poverty. The indicator commonly assumed to proxy hardship fails the available Pahang evidence. Unemployment should not be the sole district-targeting criterion.

Finding 2 — LFPR provides a stronger employment-based signal

Pearson $r = -0.564$, Spearman $\rho = -0.532$ and Kendall $\tau = -0.374$, all statistically significant; OLS $R^2 = 0.318$. Interpretation: districts with greater labour-force participation tend to have lower observed poverty. LFPR is retained as a supplementary screening indicator alongside the DVI, not as a replacement for poverty measurement.

Finding 3 — Lipis and Maran demonstrate the decoupling pattern

Lipis averages 1.83% unemployment and 8.07% poverty; Maran averages 2.13% unemployment and 7.70% poverty, compared with Pahang medians of 2.43% and 6.80%. Flagged low-unemployment/high-poverty district-years have mean LFPR 60.3% versus 63.8% for the full sample. This is consistent with a working-poor/low-participation pattern, but it is not a proven causal mechanism.

6.2 Objective 2 Findings — District Vulnerability Index

Finding 4 — DVI identifies persistent vulnerability

District	Mean DVI	Classification
Maran	0.758	High
Lipis	0.743	High
Jerantut	0.704	High
Temerloh	0.629	Moderate
Bera	0.575	Moderate
Rompin	0.545	Moderate
Raub	0.506	Moderate
Pekan	0.489	Moderate
Kuantan	0.437	Low
Bentong	0.436	Low
Cameron Highlands	0.404	Low

Table 6.1 – District's DVI score and its vulnerability classification

From 2021 onward, Maran and Lipis occupy the top two ranks every year. Jerantut and Temerloh remain relatively high, while Bentong and Cameron Highlands are near the bottom. The 2020 ranking is anomalous and should be interpreted cautiously.

Finding 5 — DVI carries significant poverty signal and improves over time

Validation	Result
Pearson correlation	$r = 0.575$, $p < 0.001$
Spearman correlation	$\rho = 0.626$, $p < 0.001$
Matched observations	$n = 33$
2020 rank agreement	$\rho = 0.31$, $p = 0.355$; mean rank gap 2.91
2022 rank agreement	$\rho = 0.61$, $p = 0.047$; mean rank gap 2.18
2024 rank agreement	$\rho = 0.64$, $p = 0.035$; mean rank gap 1.64

Table 6.2 – Validation performance for DVI

Interpretation: the DVI captures meaningful variation in actual poverty, with stronger recent-year agreement. Why it matters: the index can serve as an interim screening signal between HIES releases, provided users recognise that it is imperfect and lower-confidence in early/thinly covered years.

Finding 6 — Two broad vulnerability profiles exist

Indicator screening shows Bentong, Raub and Jerantut retaining broader sets of indicators, including demographic and basic-service variables, while the remaining eight districts are more strongly characterised by inflation and labour-market indicators. Interpretation: a single state-wide vulnerability mechanism would mask district-specific drivers. Implication: intervention type should be matched to the dominant vulnerability profile.

6.3 Objective 3 Findings — Nowcasting

Finding 7 — Income nowcasting is materially more reliable than poverty nowcasting

Income achieves LOOCV MAE RM 278.45, RMSE RM 356.26, MAPE 6.31% and R^2 0.662. Poverty achieves MAE 1.80 pp, RMSE 2.17 pp, MAPE 48.07% and R^2 0.533. The dashboard should communicate these outcomes with different confidence levels.

Finding 8 — Poverty model selection involves an explicit trade-off

The unrestricted Decision Tree achieved poverty $R^2 = 0.635$ versus Ridge $R^2 = 0.533$. Ridge was nevertheless retained for stability and interpretability. This is a deliberate interpretability-over-accuracy decision and should be stated whenever poverty nowcasts are presented.

Finding 9 — Nowcast suggests persistent high-poverty districts through 2027

The Pahang-wide mean nowcast poverty rises from 7.36% in 2025 to 7.87% in 2027. Rompin is projected from 12.02% to 12.52%; Lipis from 9.55% to 10.51%; Bera from 8.90% to 9.71%; and Pekan from 8.79% to 9.69%. Bentong remains comparatively flat at 5.78% to 5.75%. These trajectories suggest persistence rather than convergence, but exact values are indicative rather than precise forecasts.

Finding 10 — Some income nowcasts repeat because of tree structure

Pahang-wide mean projected income is nearly flat at RM 4,560.84 in 2025 and RM 4,600.70 in 2026–2027. Jerantut, Pekan, Raub, Rompin and Temerloh show identical or near-identical values across multiple projected years. This is a direct artifact of shared depth-3 tree leaves and should not be narrated as genuine income stagnation.

Finding 11 — Poverty prediction intervals are wide and cannot yet be validated

For example, Rompin's 2027 poverty interval spans 10.69%–18.57%, approximately 7.9 percentage points around a point estimate of 12.52%. Kuantan's 2027 income interval is RM 4,631–RM 5,779 around RM 5,432. Future interval coverage is not yet testable because 2025–2027 outcomes are unobserved.

6.4 AI Integration - Clustering Finding

Finding 12 — Three robust typologies exist, confirmed by two independent algorithms (ARI = 1.000)

Labour/Household-driven (Bera, Jerantut, Maran, Raub, Rompin, Temerloh); Living-condition-driven (Bentong, Cameron Highlands, Kuantan, Pekan); Compound labour + living-condition (Lipis, alone).

Finding 13 — Lipis has a compound vulnerability the DVI ranking hides

It is the only district elevated on two dimensions simultaneously, yet ranks only mid-table on the composite DVI.

Finding 14 — Typology and DVI severity are complementary

Cluster 0 alone spans High, Moderate, and Low DVI-ranked districts — clustering answers "what kind of help," DVI answers "how urgent."

Finding 15 — Data confidence varies by district

Maran's Labour, Household, and Living-Condition dimensions were all screened out, leaving only a non-discriminating Macro dimension. Its top DVI rank should be read with this in mind.

6.5 Prescriptive Findings

The findings suggest that socioeconomic assistance in Pahang should move beyond unemployment-based targeting and adopt a multidimensional approach. The DVI, LFPR and nowcast indicators can be combined to identify districts requiring closer attention, with Bera, Pekan, Jerantut, Lipis and Rompin emerging as priority areas for 2025–2027 monitoring. Intervention strategies should also reflect each district's dominant vulnerability profile rather than applying a uniform policy. However, nowcast estimates should be treated as early-warning decision-support signals, not replacements for official HIES statistics, and should be presented together with their uncertainty intervals.

6.6 Consolidated Findings → Response

Finding	Evidence	Decision response
Unemployment–poverty decoupling	$r \approx -0.02$ to -0.04 ; $p > 0.8$	Stop unemployment-only targeting
LFPR is stronger	$r = -0.564$; $R^2 = 0.318$	Use LFPR as supplementary screen
Lipis/Maran pattern	Low unemployment + high poverty + low LFPR	Target for further assessment
DVI validity	$r = 0.575$; $p = 0.626$	Use DVI as interim screening signal
District profiles differ	Indicator screening differs by district	Tailor intervention type
Income nowcast stronger	R^2 0.662; MAPE 6.31%	Use income with higher confidence
Poverty nowcast weaker	R^2 0.533; MAPE 48.07%	Use as directional early warning only
High-risk watch-list	Rompin/Lipis/Bera/Pekan persist	Prioritise monitoring before next HIES

Table 6.3 — Summary of major findings, Objectives 1, 2 and 3

7. Proposed Data / AI Solution

7.1 Solution Concept

The proposed solution, NADA (New-era Assistance & Distribution Agility), is a decision-support system that turns fragmented district-level socioeconomic data: labour, inflation, agriculture, amenities and demography, into a timely, transparent and actionable picture of economic vulnerability across Pahang's 11 districts. It will be used by UPEN Pahang, district councils and NGOs to identify which districts most need development funds,

infrastructure investment or targeted aid in a given year, and which districts share the same underlying cause and could be served by a shared intervention design rather than eleven separate ones.

Inputs are the DOSM/OpenDOSM datasets described in Section 2; outputs are the validated DVI score and ranking (Section 6.2), a nowcasted estimate of median income/poverty for non-survey years (Objective 3), and a district typology layer that groups districts by shared vulnerability type and diagnoses each district's most disproportionately elevated dimension (Section 5.7). The decision it supports is not only where to prioritise scarce state and NGO resources between HIES releases, but which districts can be grouped into one intervention and what that intervention should target.

7.2 Solution Architecture / Workflow

The technical architecture is: Python (pandas, numpy, scikit-learn, statsmodels, scipy) for ingestion, cleaning, indicator construction, statistical modelling and the clustering layer; Supabase PostgreSQL as the single structured data layer feeding the Power BI dashboard via DirectQuery. The clustering module (K-Means, validated against Agglomerative/Ward hierarchical clustering) runs downstream of the DVI dimension-scoring pipeline, writing typology assignments, cluster profiles, and confidence flags back into the same verified database layer — so the dashboard's typology page draws from the same single source of truth as the DVI ranking and nowcasting pages, with no separate or unverified data path.

7.3 Innovation

- Inter-survey economic nowcasting provides an interim evidence layer instead of waiting for the next HIES release.
- Multidimensional vulnerability combines labour, living conditions, demographic pressure and price pressure rather than relying on unemployment alone.
- Model complexity is deliberately matched to the small panel; transparency and explainability are treated as design strengths.
- An unsupervised district typology layer groups the 11 districts by shared vulnerability type using K-Means clustering, independently confirmed by Ward hierarchical clustering (Adjusted Rand Index = 1.000) — moving the solution beyond a single severity ranking to answer "which districts need the same kind of help, and why," while explicitly flagging districts where indicator coverage is too thin to trust (e.g. Maran).

8. Interactive Dashboard

8.1 Dashboard Objective

The NADA Power BI dashboard translates the project's statistical, vulnerability, clustering and nowcasting analyses into an interactive decision-support tool for UPEN Pahang, district councils and NGOs. It enables users to identify districts requiring attention, understand their dominant vulnerability characteristics, and monitor estimated income and poverty between HIES release years. The dashboard is intended as a complement to official statistics, not a replacement for HIES.

8.2 Dashboard Users

- Primary Users: UPEN Pahang state planners and economists responsible for regional development and welfare resource allocation.

- **Secondary Users:** District councils and NGOs requiring district-level vulnerability insights for targeted aid and local interventions.
- **Key Use Cases:** Monitoring district vulnerability, identifying potential working-poor areas, comparing district typologies and vulnerability drivers, and supporting inter-survey income and poverty monitoring for evidence-based resource prioritisation.

8.3 Dashboard Structure

The dashboard consists of three interconnected pages aligned with the project's analytical objectives:

- **Page 1: Executive Overview & Labour Mismatch (Objective 1):** Evaluates headline macroeconomic indicators and tests the decoupling hypothesis between unemployment rates and poverty incidence across all districts.
- **Page 2: District Vulnerability Index & Typology (Objective 2):** Ranks districts using the DVI, decomposes vulnerability into its underlying dimensions, and uses clustering analysis to group districts with similar vulnerability profiles. This supports more targeted intervention by showing that districts may experience vulnerability through different combinations of labour-market, demographic, living-condition and economic factors.
- **Page 3: Economic Nowcasting & Uncertainty Simulation (Objective 3):** Presents estimated present-day median income and poverty trajectories during non-survey gap years, complete with explicit 95% confidence intervals and LOOCV validation metrics.

8.4 Key Performance Indicators

KPI	Definition	Why It Matters	User Decision Supported
Highest Risk District & DVI Score	District with the highest composite DVI score, e.g. Maran (0.758)	Identifies where multiple vulnerability factors are most concentrated	Prioritise high-urgency welfare and economic interventions
Unemployment vs. Poverty Correlation (ρ)	Spearman correlation between unemployment and poverty: $\rho = -0.04$, $p = 0.836$	Shows that low unemployment does not necessarily indicate low poverty	Prevent unemployment-only targeting
State Average Poverty Rate & LFPR	Poverty incidence 6.22% and LFPR 63.85%	Provides a benchmark for evaluating district performance	Set state-wide and district benchmark thresholds
Nowcasted Poverty & Median Income	Estimated poverty (%) and median income (RM), with 95% prediction intervals	Fills the information gap between HIES releases	Support annual planning and budget review in non-survey years
District Vulnerability Typology	Groups districts with similar socioeconomic and vulnerability profiles using clustering analysis.	Shows that districts can experience vulnerability through different combinations of factors.	Match interventions to district type and vulnerability profile.

Dominant Vulnerability Driver	Identifies the dimension or factors contributing most strongly to a district's vulnerability profile.	Helps explain why a district is vulnerable rather than relying only on its overall DVI score.	Select more targeted interventions and resource allocation.
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Table 8.1 – Key performance indicator used in dashboard

8.5 Filters

- District: Slicer options covering Pahang's 11 administrative districts (Bentong, Bera, Cameron Highlands, Jerantut, Kuantan, Lipis, Maran, Pekan, Raub, Rompin, Temerloh).
- Year: Temporal selector spanning available historical and nowcasted periods (2019, 2020, 2021, 2022, 2023, 2024).
- Vulnerability Level: Categorical risk filter (High, Moderate, Low).

8.6 Dashboard Screenshots

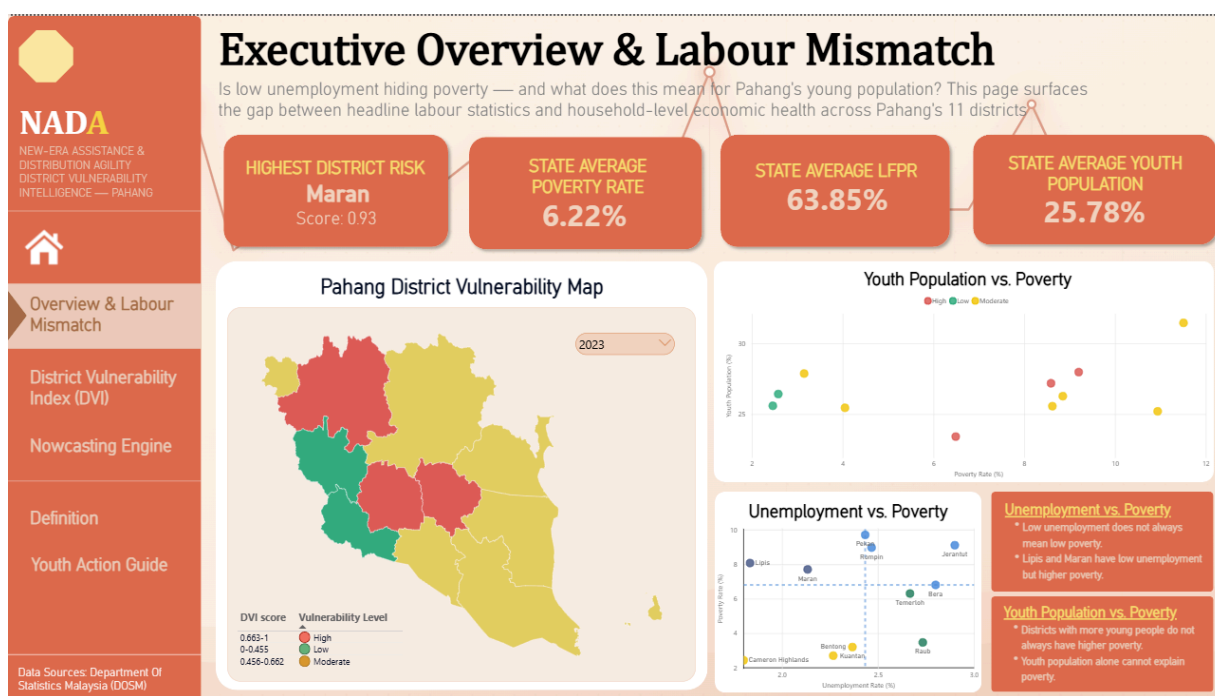


Figure 8.1 – Executive Overview & Labour Mismatch page

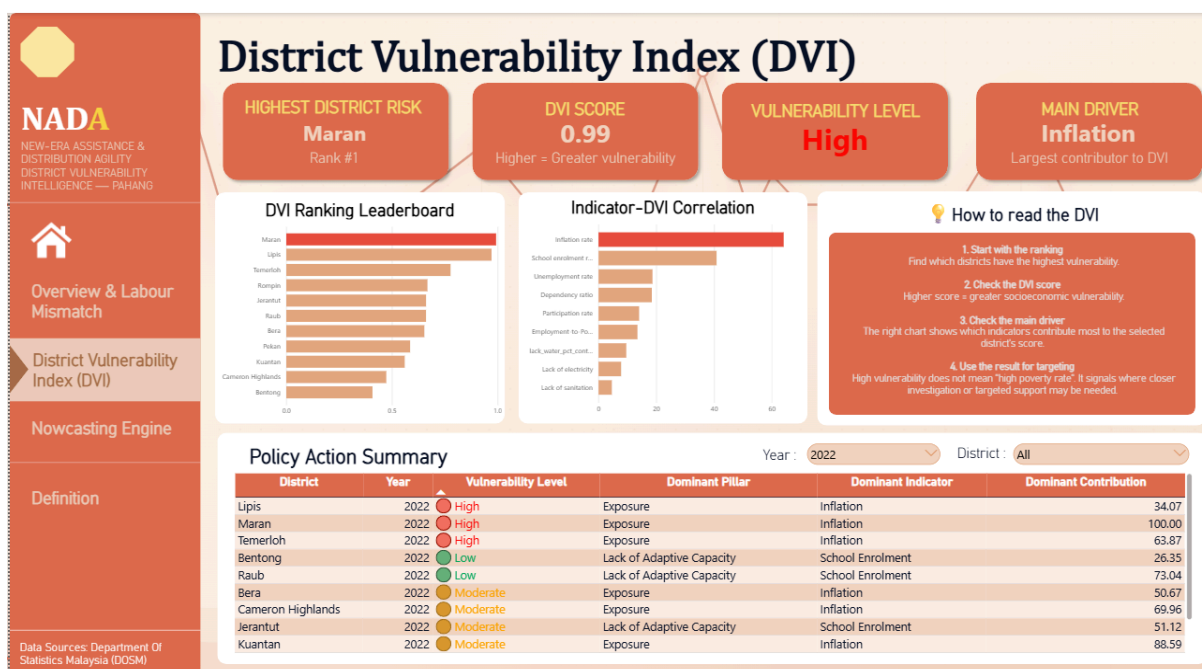


Figure 8.2 – District Vulnerability Index page

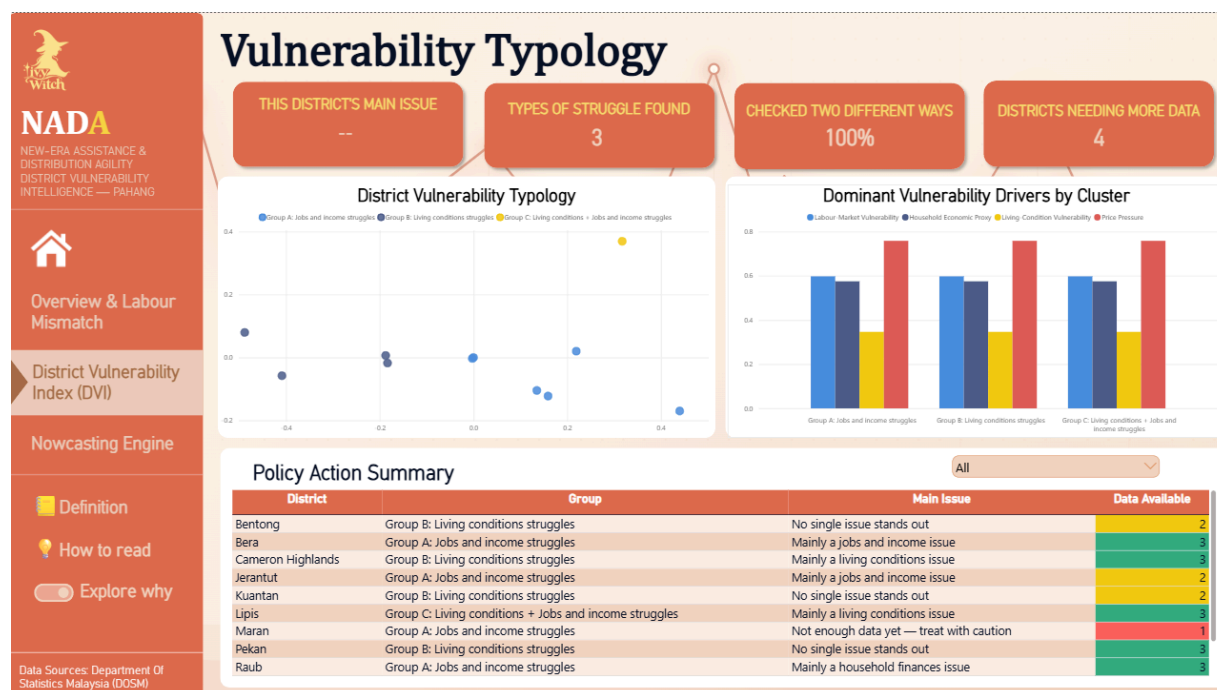


Figure 8.3 - Subpage District Vulnerability Page (Vulnerability Topology)

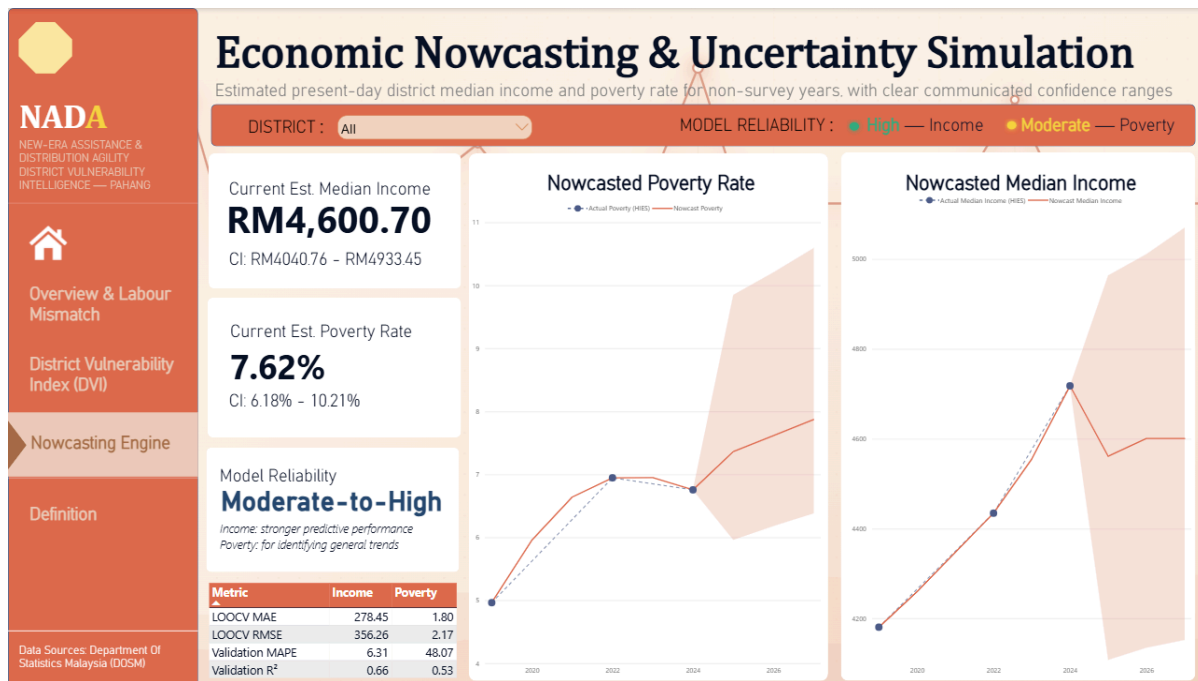


Figure 8.4 – Economic Nowcasting & Uncertainty Simulation page

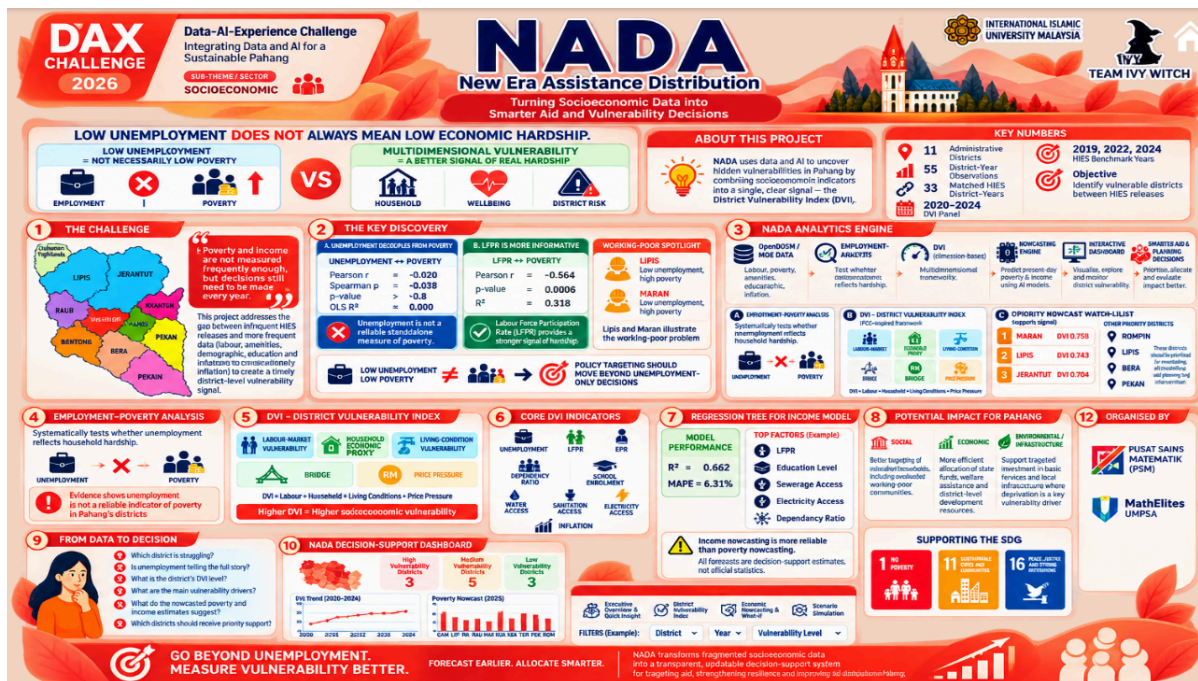


Figure 8.5 - Infographic page

8.7 Dashboard-to-Decision Mapping

Output	Evidence basis	Stakeholder	Decision use
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Potential Working-Poor Zone	Unemployment–poverty decoupling; Lipis/Maran examples	UPEN Pahang	Reassess unemployment-only targeting
DVI priority ranking	Validated multidimensional vulnerability signal	UPEN + councils + NGOs	District prioritisation
Dominant vulnerability driver	District-level indicator screening / DVI decomposition	District councils / agencies	Match intervention to driver
Income nowcast	LOOCV R^2 0.662; MAPE 6.31%	Budget/planning teams	Monitor inter-survey income
Poverty nowcast watch-list	LOOCV R^2 0.533; MAPE 48.07%; wide intervals	UPEN + councils	Early-warning monitoring, not automatic funding
Observed vs nowcast comparison	HIES anchors and subsequent releases	Analysis team / UPEN	Validation and recalibration
District Vulnerability Typology	Clustering analysis groups districts with similar socioeconomic and vulnerability profiles	UPEN Pahang; district councils; NGOs	Match intervention strategies to similar district profiles

Table 8.2 – Evidence of the insights shown in the dashboard and how it would help in decision-making.

9. Recommendations

Recommendation	Evidence	Stakeholder	Timeline	KPI
Stop unemployment-only targeting	Finding 1	UPEN Pahang	Short term	Allocation vs. DVI/poverty correlation
Prioritise Bera, Lipis, Maran	Finding 4	UPEN Pahang; district councils	Short term	YoY DVI trend, top-3 districts
Differentiate intervention type by cluster	Finding 6	District councils; state agencies	Medium term	Dimension-level DVI movement
Caution on 2020 rankings and Maran/Kuantan scores	Finding 5; Section 5.5	UPEN Pahang; analysis team	Short term	Confidence flag adoption in dashboard
Nowcasts as monitoring tool, not sole trigger	Findings 7, 8, 11	Analysis team; UPEN Pahang	Short term	Calibration coverage rate
Prioritise Rompin, Lipis, Bera, Pekan (nowcast watch-list)	Finding 9	UPEN Pahang; district councils	Short term	Nowcast trend vs. next HIES release

Table 9.1 — Summary of recommendations

10. Potential Impact for Pahang

10.1 Social Impact

A validated, more-frequent vulnerability signal could help direct welfare distribution, food-bank assistance and community support toward the households and districts (proposed) most likely to be genuinely struggling, including working-poor households that unemployment-based criteria currently overlook — most notably in Lipis and Maran (Finding 3).

10.2 Economic Impact

By separating districts that need job creation (genuinely high unemployment) from districts that need wage support or infrastructure investment (working-poverty zones), state and district resources could (proposed) be allocated with less waste than under a uniform, unemployment-only policy.

10.3 Environmental / Sustainability Impact

The DVI's living-condition dimension (water, sanitation, electricity access) could (proposed) inform locally targeted infrastructure and resource-access investment in the districts where it is the dominant driver (Bentong, Raub, Jerantut), supporting Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) and Goal 11 (Sustainable Cities and Communities).

10.4 Operational / Governance Impact

A dashboard-delivered, annually updatable DVI could (proposed) give UPEN Pahang, district councils and NGOs a continuously monitorable vulnerability signal in place of waiting for the next HIES release, enabling earlier, more evidence-based planning and supporting Goal 16 (Peace, Justice and Strong Institutions) through a transparent, fully decomposable index rather than an opaque score.

10.5 Impact Measurement Framework

Desired Impact	Indicator	Baseline	Target	Measurement Method
Reduced unemployment-based misallocation	Share of district-level aid decisions using unemployment as sole criterion	Current practice (unemployment-based)	DVI/LFPR-informed criteria adopted	Policy/process review with UPEN Pahang
Improved targeting of persistently vulnerable districts	DVI rank stability for Bera, Lipis, Maran	2021–2024 baseline rank (top 3)	Sustained or improved rank alongside targeted intervention	Annual DVI recalculation
Faster response between HIES rounds	Time between a vulnerability change and a policy response	Multi-year HIES cycle only	Annual DVI-informed review cycle	Dashboard usage / decision log

Table 10.1 – Framework on the impact of NADA project framework

11. Implementation Roadmap

Phase	Activity	Stakeholder	Output
1. Validation completion	Complete nowcast validation/calibration/residual diagnostics and integrate validated outputs	Analysis team	Validated model or documented non-viability
2. Dashboard build	Connect Power BI to DVI and nowcasting data layer	Analysis team + UPEN IT	Working dashboard prototype
3. Stakeholder pilot	Test DVI ranking and dashboard with UPEN and 1–2 district councils	UPEN + councils	Usability/decision-fit feedback
4. Refinement	Add confidence/coverage flags and incorporate feedback	Analysis team	Refined dashboard/documentation
5. Monitoring	Recalculate DVI as new indicator years arrive	UPEN + analysis team	Annual DVI ranking

Table 11.1 — Proposed implementation roadmap beyond the competition.

12. Limitations and Future Work

- Validation relies on only three HIES release years; the 2019→2020 match introduces a one-year lag.
- Coverage gaps exclude health, direct income/poverty, social protection, agriculture dependence and hazard exposure from the DVI.
- District-specific indicator screening creates formula heterogeneity, especially for Maran and Kuantan.
- State-level CPI is applied uniformly and may understate local price variation.
- Three of six historical nowcasting years are interpolated, creating circularity risk.
- Future predictors are linearly extrapolated and cannot capture shocks or turning points.
- Prediction intervals for 2025–2027 cannot yet be empirically calibrated.
- Results are specific to Pahang's 11 districts and should be re-estimated before transfer elsewhere.
- Future work should improve poverty predictors, investigate the working-poor mechanism, and refine tree resolution if year-to-year income detail is required.

13. Ethical, Responsible AI and Data Governance

NADA is designed and communicated as a decision-support system, not a substitute for official statistics, and its limitations are stated explicitly rather than minimised. The project does not, and will not:

- claim that unemployment causes poverty, or otherwise treat correlation as causation;
- classify individual households as poor using district-level signals alone;
- present the DVI as an official poverty rate, or any future nowcasts/simulations as official DOSM statistics or forecasts;
- claim exact poverty prediction where the data cannot support that claim;
- fabricate data, invent statistical results, fabricate references, or attempt re-identification of individuals.

The clustering module uses only district-level aggregate statistics. No personal or household-level data used. When indicator screening leaves insufficient evidence for a district, the system reports this explicitly ("insufficient data") rather than presenting a fabricated result, and imputed values are mathematically prevented from being mistaken for genuine findings ($z = 0$ by construction). Equal weighting across dimensions avoids introducing arbitrary bias into the typology.

Planned responsible-AI safeguards for the full system include: labelling every dashboard value as Official/Observed, NADA Estimated/Nowcasted, or Scenario/What-if Result; displaying source, year and geographic coverage for every indicator shown; storing model version and calculation metadata for auditability; providing dimension-level explanatory factors behind every DVI ranking rather than an opaque score (Section 5.6).

13.1 Generative AI Disclosure

Generative AI tools were used to assist with drafting and structuring this technical report from the team's own analysis outputs, methodology notes and proposal materials, and with language editing. All analytical results, statistical outputs, figures and interpretations presented in this report were produced by the team's own data pipeline and are reported here as supplied by the team; the team remains responsible for all methodology, analysis, findings and conclusions presented in this report.

14. Conclusion

This report addressed the first three objectives of the DAX 2026 Economy (DOSM Pahang) sub-theme. On Objective 1, pooled correlation, clustered bootstrap and OLS regression show unemployment rate has no statistically reliable relationship with household poverty across Pahang's 11 districts, while Labour Force Participation Rate does, explaining roughly a third of cross-district variance. Lipis and Maran provide named evidence of this: below-median unemployment with above-median poverty and below-average participation, consistent with a working-poor pattern rather than genuine labour-market health.

On Objective 2, a District Vulnerability Index built from indicators already published annually or monthly was constructed and validated three independent ways, correlating moderately and significantly with actual poverty ($r = 0.567$, $n = 33$) and improving in reliability from unreliable in 2020 to good by 2024. Bera, Lipis and Maran emerge as the most persistently vulnerable districts. An unsupervised clustering layer extends this from a single severity ranking into shared vulnerability typologies, revealing that Lipis carries a compound labour-and-living-condition risk the DVI ranking alone does not surface, and flagging districts such as Maran where indicator coverage is too thin for confident conclusions.

On Objective 3, a nowcasting engine built on a PCHIP-completed six-year panel and validated by leave-one-out cross-validation estimates median household income more reliably ($R^2 = 0.66$, $\text{MAPE} = 6.31\%$) than poverty ($R^2 = 0.53$, $\text{MAPE} = 48.07\%$), with Ridge retained over a better-scoring but less transparent Decision Tree for interpretability. Its 2025–2027 projections show Pahang's higher-poverty districts. Rompin, Lipis, Bera and Pekan are persisting and mildly worsening rather than converging toward the state average; poverty's wider prediction intervals mean the engine is best used as an early-warning monitor rather than a precise forecast.

The analytical contribution of this work is a validated, decomposable, updatable alternative to unemployment-based targeting that also identifies which districts share a common vulnerability type. NADA delivers this through a three-page Power BI dashboard combining the validated DVI, nowcasting engine and district typology layer (Sections 5–6). Better-supported decisions this analysis can already inform include which districts to prioritise for development funds and aid, and which districts can be grouped under one shared intervention design. Realising the full impact described in Section 10 depends on putting these validated outputs in front of UPEN Pahang, district councils and NGOs for real decision-making use.

15. References

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Appendices

Appendix A: Full Data Dictionary

Table A.1 — Extended fields (beyond Table 2.2)

Raw: students_total, population_5_19 (enrolment inputs); piped_water_pct, sanitation_pct, electricity_pct (pre-reversal access shares); labour_force_thousand, employed_population_thousand (LFPR/EPR inputs); crop_production_tonnes (explored, excluded per Table 1.1 scope).

Derived — DVI: L, H, C, P (four dimension scores, 0–1); DVI_dimension_based, DVI_ipcc; vulnerability_level; obs_interpolated_flag.

Derived — Clustering (Section 5.7): cluster, cluster_label, cluster_plain_name, n_dimensions_available, dimensions_imputed, dominant_dimension, dominant_dimension_zscore, plain_status, recommended_focus, PC1, PC2.

Appendix B — Additional Statistical Results

Table B.1 — k-selection (Section 5.7.2)

k	Silhouette
2	0.427
3	0.392
4	0.375
5	0.368

Table B.2 — Cluster vs. Vulnerability Level (Finding 14)

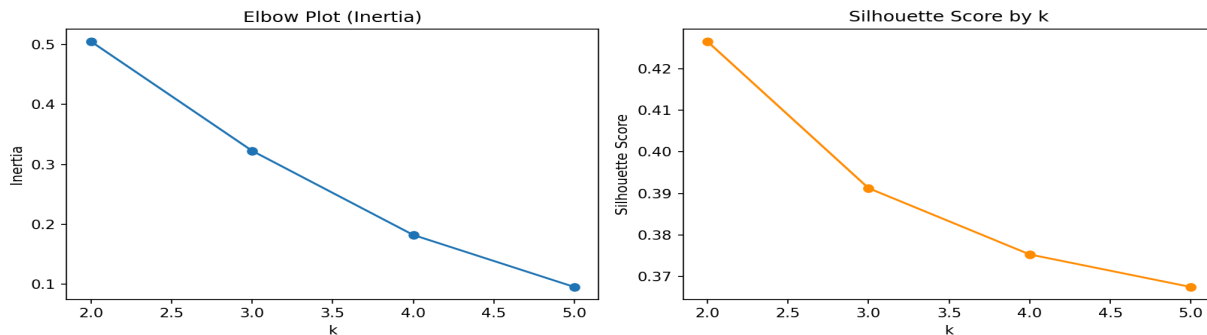
Cluster	High	Moderate	Low
0 - Labour-Market	1	4	1
1 - Living Condition	0	3	1
2 - Compound	0	1	0

Appendix C — Model Parameters

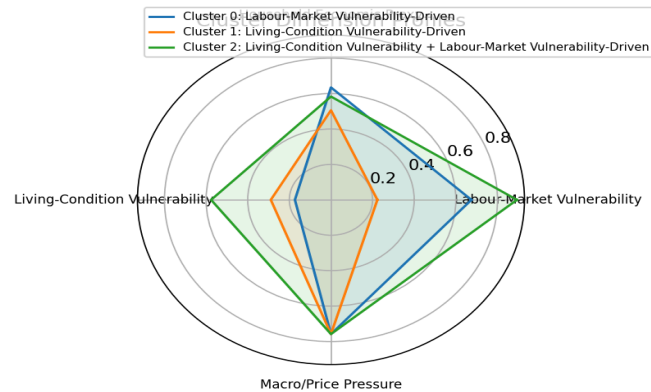
DVI/nowcasting: as in Sections 5.4–5.6. Clustering: K-Means (n_clusters=3, n_init=50, random_state=42); Ward hierarchical validation (n_clusters=3, linkage='ward'); z-score diagnosis computed pre-imputation, imputed/zero-variance dimensions excluded from ranking; PCA (n_components=2, random_state=42) — PC1 75.1%, PC2 17.6%, combined 92.6%.

Appendix D — Supplementary Visualisations

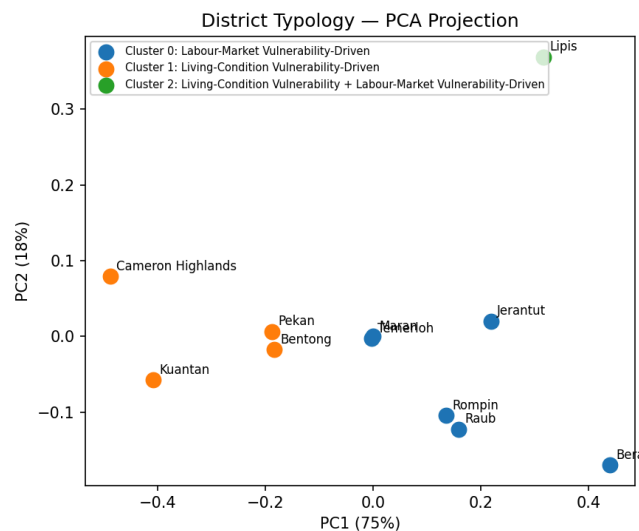
Elbow/silhouette plot (k=2–5)



Cluster centroid radar plot



PCA scatter of the three typologies



Appendix E — Dashboard Link

<https://app.powerbi.com/view?r=eyJrIjoiaMTBhMGQ5MjMtY2QwMC00NDkwLTg1N2UtNDY5YmVjNDVhYTkxliwidCI6IjE4Y2U3NmY2LTk5ZjQtNDU3Zi05ZjYyLWFiZDY1ZDliOTc3NyIsImMiOiJEWfQ%3D%3D>

Appendix F — Reproducibility Information

Python 3.x; pandas, numpy, scikit-learn, scipy, statsmodels, matplotlib. random_state=42 throughout clustering (K-Means, PCA); nowcasting bootstrap uses 5,000 resamples (Section 5.6.7).

Appendix G — Generative AI Disclosure

Generative AI (Claude, ChatGPT, Gemini) was used to help in contributing the code for EDA and analytical tasks. All results were produced by the team's own pipeline and reviewed by the team, who remain responsible for all methodology, analysis, and conclusions.